Amazon Web Services (AWS) Overview

Topics

- Brief history of AWS
- The building blocks of AWS
- Who owns what in the cloud?
- Compute, Storage, Database, and Networking
- What is the well-architected framework

Brief History of AWS

Brief History of AWS

2003

Pinkham & Black suggested selling amazon's IaaS

2004

SQS officially launched

2006

AWS officially launched (EC2 and S3); targeted startups

2007

Over 180K developers on the platform

2010

amazon.com moved over

2012

First re:Invent Conference

Brief History of AWS

2013
Certs Launched (Associate, Professional, and Specialty)

2015
AWS breaks out its revenue: $6 Billion USD annually

2016
Run rate of $13 Billion USD

2017
AI services; run rate of $27 Billion USD

2019
ML Specialty Cert launches

2020
Run rate of $40 Billion USD

2023
Run rate of $90.8 Billion USD

2024
Run rate of $100 Billion USD

The Building Blocks of AWS

AWS Global Infrastructure Map

34 Launched Regions with multiple Availability Zones (AZs); 108 AZs; AZ = Data Center(s)

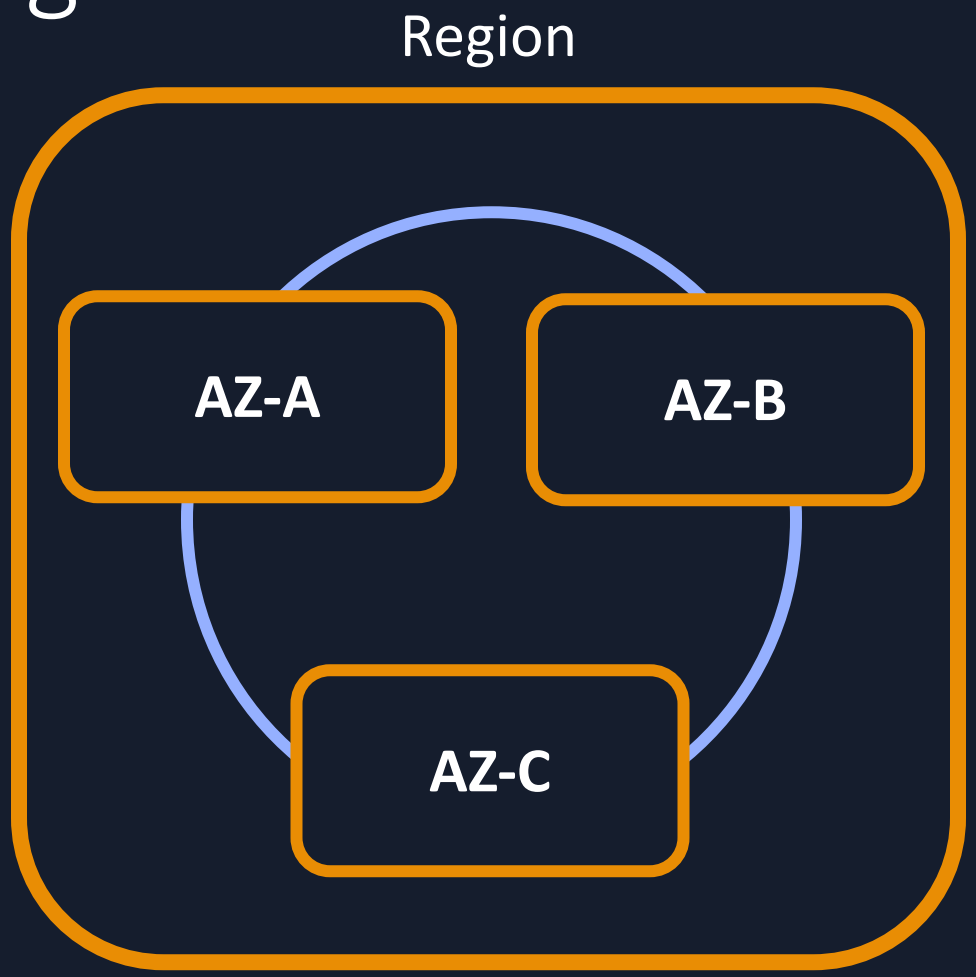
Regions Coming soon

https://aws.amazon.com/about-aws/global-infrastructure/

What is an Availability Zone (AZ)?

- An AZ may be several data centers - counted as 1 AZ when they are a couple of miles of each other

What is a Region?

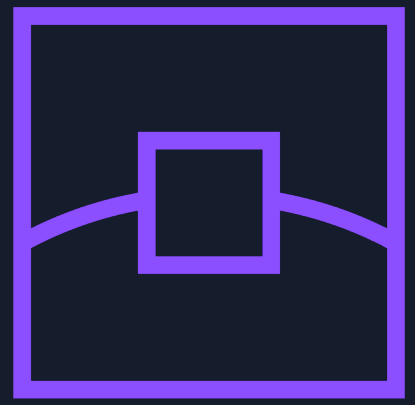


A Region is a geographical area. Each region consists of two (or more) AZs.

https://aws.amazon.com/about-aws/global-infrastructure/regions_az/?p=ngi&loc=2

What are CloudFront's POPs (points of presence)?

- CloudFront's points of presence (POPs, also known as edge locations) are used for caching content
- CloudFront has two tiers of edge locations: POPs and regional edge caches
- POPs are more numerous and closer to users, but they have smaller caches
- Currently 600+ CloudFront POPs and 13 Regional edge caches



Edge Location

Amazon CloudFront

Checkpoint

The AWS Global Infrastructure comprises Regions, Availability Zones (AZs), and Edge Locations, and there is a different number of each infrastructure element. Select the option that shows the correct order from greatest to least.

1. Number of Edge Locations > Number of AZs > Number of Regions
2. Number of Regions > Number of AZs > Number of Edge Locations
3. Number of AZs > Number of Edge Locations > Number of Regions
4. Number of AZs > Number of Regions > Number of Edge Locations

What does an AWS Region consist of?

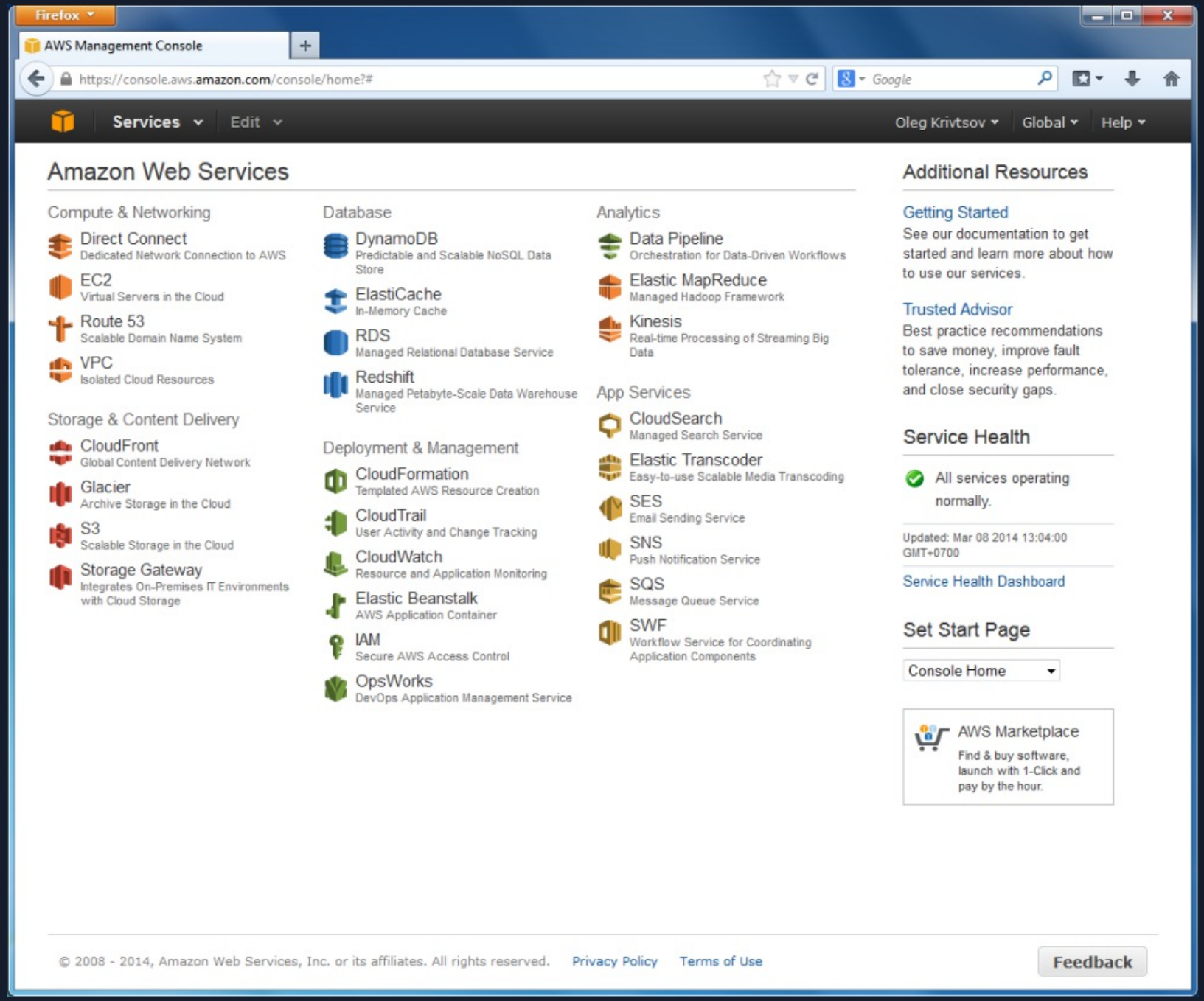
1. A collection of databases that can only be accessed from a specific geographic region.
2. A collection of data centers that is spread evenly around a specific continent.
3. A console that gives you a quick, global picture of your cloud computing environment.
4. Each AWS Region consists of multiple, isolated, and physically separate Availability Zones within a geographic area.

AWS Service Types

		Front-end Web & Mobile
End User Computing	Quantum Technologies	Containers
IoT	Customer Enablement	Game Development
Customer Engagement	Business Applications	Desktop & App Streaming
AR & VR	Application Integrations	AWS Cost Management
Analytics	Security, Identity, & Compliance	Mobile
Management & Governance	Media Services	Machine Learning
Robotics	Blockchain	Satellite
Migration & Transfer	Network & Content Delivery	Developer Tools
Compute	Storage	Databases

AWS Global Infrastructure

2014 AWS Console



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AWS Global Infrastructure		

Who owns what in the Cloud?

Renting a Car

- When you rent a car, who's responsible for what?

You

- Not to damage it
- Not to speed
- Pay tolls, etc.

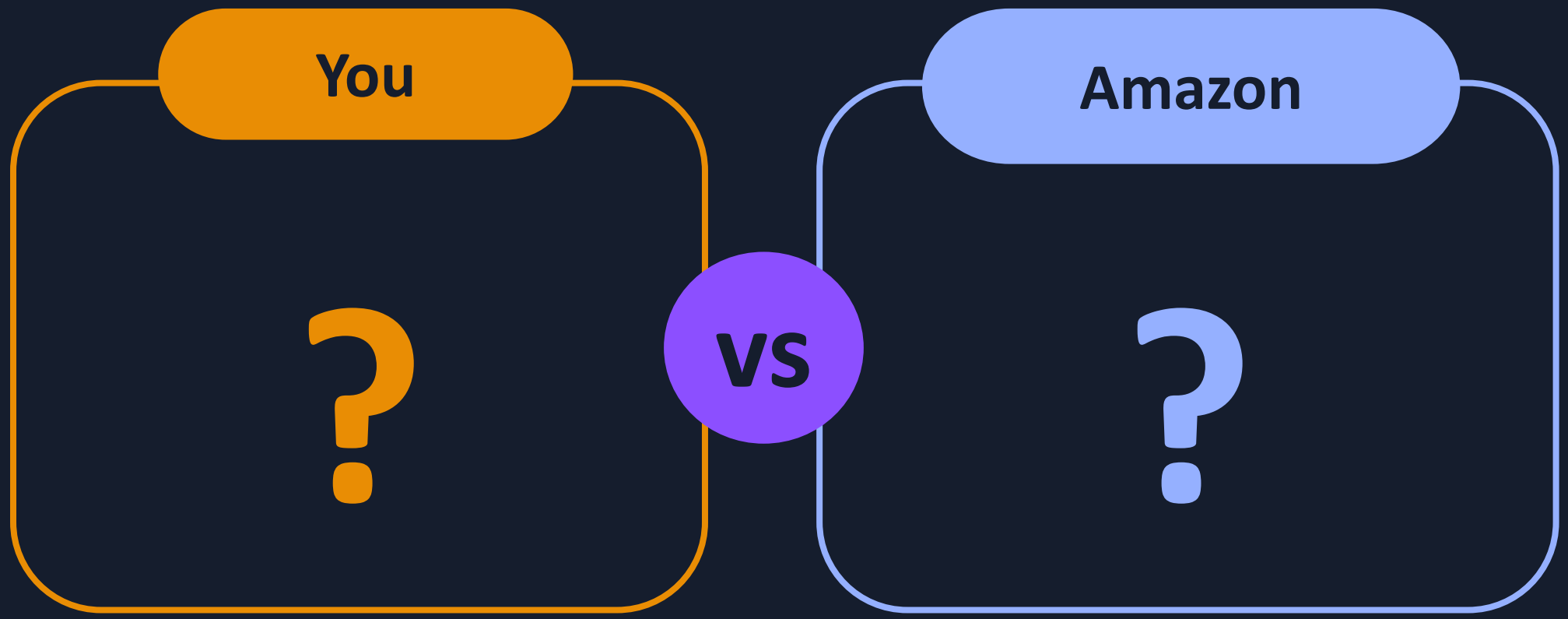
VS

Rental Company

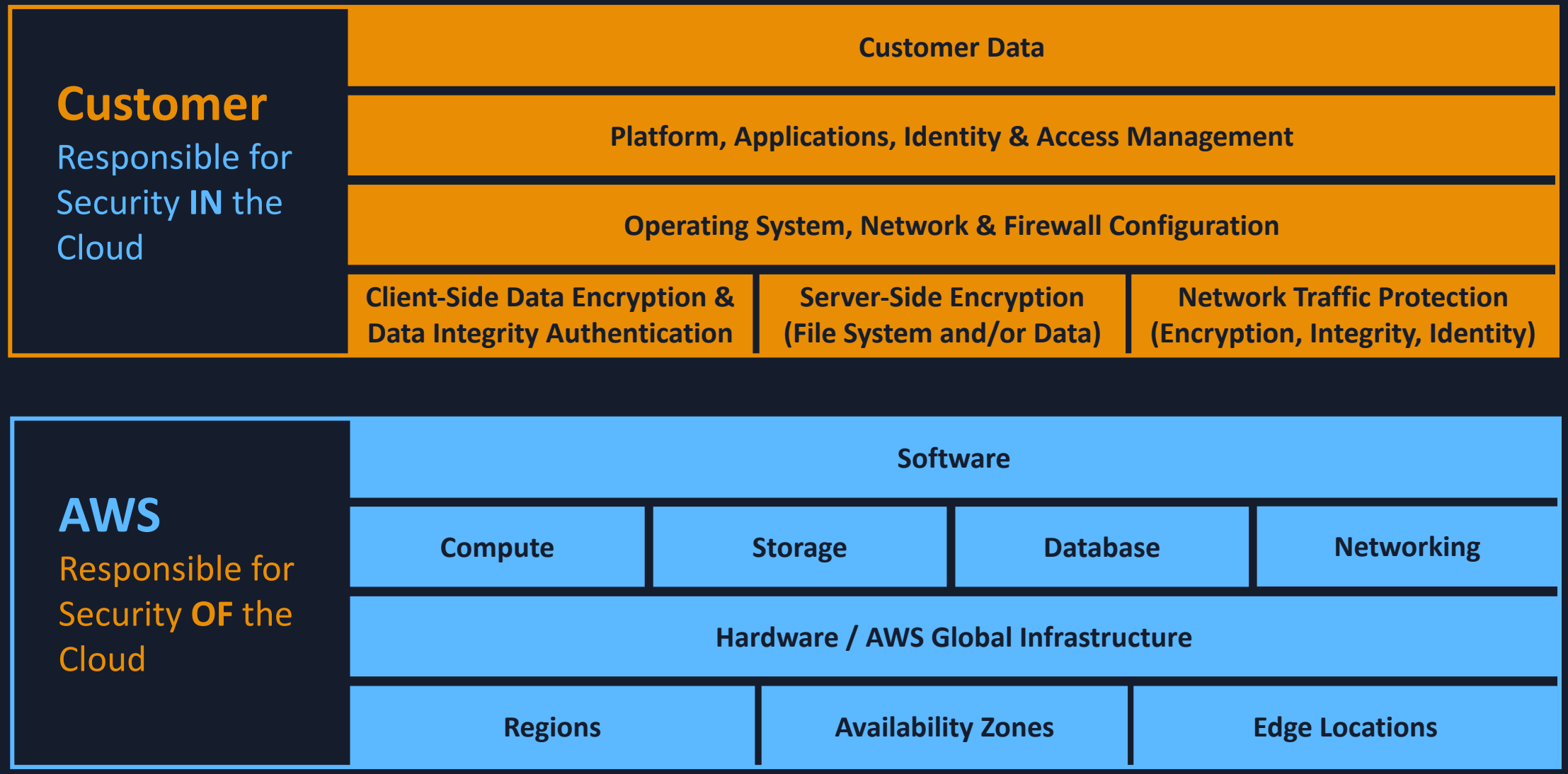
- Tire pressure
- Full tank of gas
- Mechanics

Renting Resources in the Cloud

- Who’s responsible for what when you rent resources in the cloud?



The Shared Responsibility Model



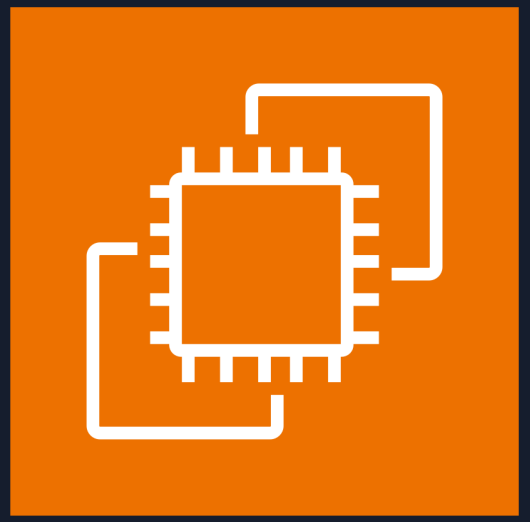
Can you do this yourself in the AWS Management Console?

- If yes, you are likely responsible.
  - Things you can control
  - Security groups, IAM users, patching your EC2 operating systems, patching your databases running on EC2, etc.
- If no, AWS is likely responsible.
  - Things you can’t control
  - Management of data centers, security cameras, cabling, patching RDS, operating systems, and so on
- Encryption is a shared responsibility.

Compute, Storage, Database, and Networking

Compute

- You wouldn’t be able to build an application without compute power



Amazon Elastic Compute Cloud (Amazon EC2)

AWS Lambda

AWS Elastic Beanstalk

Storage

- Think of storage as a giant disk in the cloud

Amazon Simple Storage Service (Amazon S3)

Amazon Elastic Block Store (Amazon EBS)

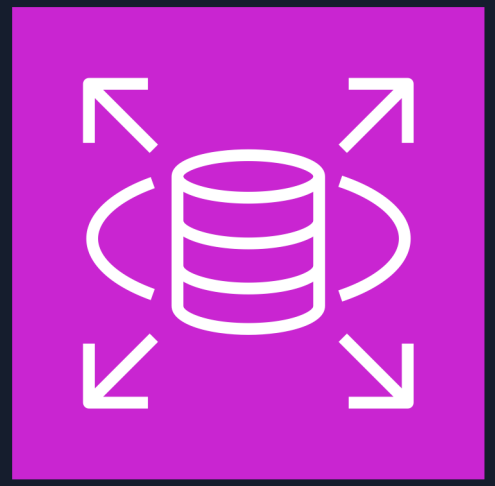
Amazon Elastic File System (Amazon EFS)

AWS Storage Gateway

Amazon FSx

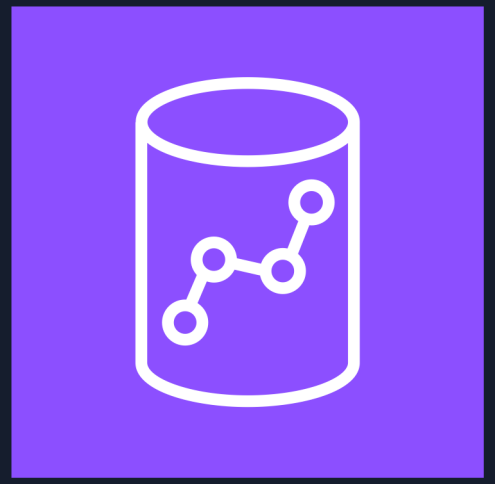
Databases

- The easiest way to think of a database is simply a spreadsheet



Amazon Relational Database Service (Amazon RDS)

Amazon DynamoDB



Amazon Redshift

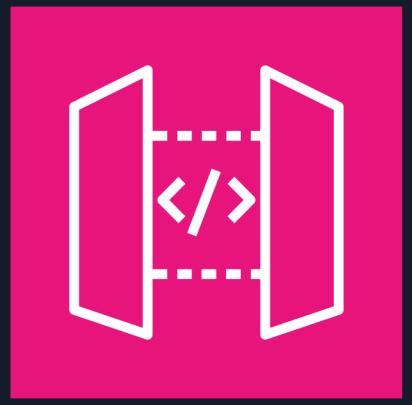
Networking

- We need a way for our compute, storage, and databases to communicate and even a place for them to live

Amazon Virtual Private Cloud (Amazon VPC)

AWS Direct Connect

Amazon Route 53



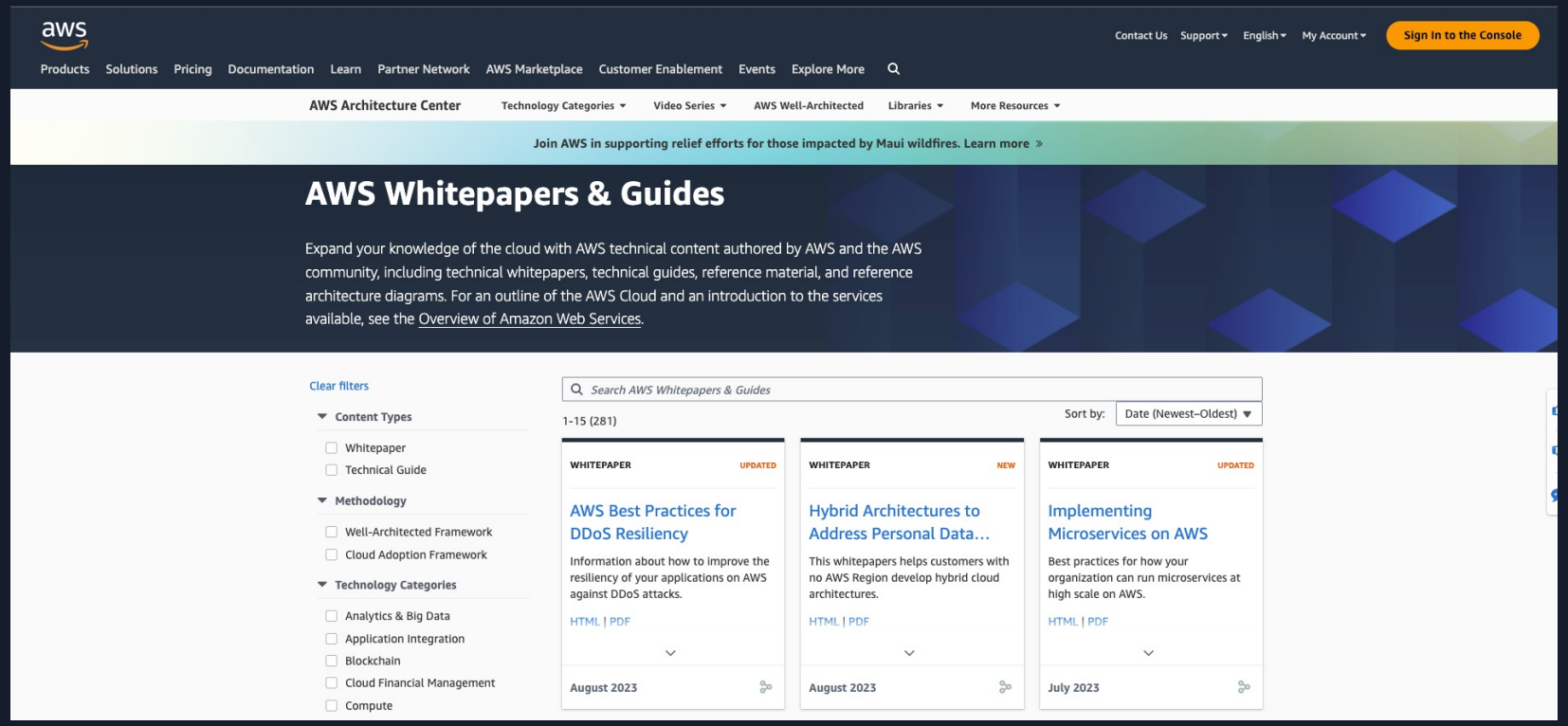
Amazon API Gateway

AWS Global Accelerator

What is the Well-Architected Framework?

AWS Whitepapers

- AWS has hundreds of whitepapers available
- https://aws.amazon.com/whitepapers



The Pillars of the AWS Well-Architected Framework

Operational Excellence
Running and monitoring systems to deliver business value, and continually improving processes and procedures.

Security
Protecting data, systems, and assets in a way that can improve security posture.

Reliability
Ensuring a workload performs its intended function correctly and consistently when it’s expected to.

Performance Efficiency
Use computing resources efficiently to meet system requirements, and to maintain that efficiency as demand changes and technologies evolve.

Cost Optimization
Run systems to deliver business value at the lowest price point and avoiding unnecessary costs.

Sustainability
Reducing energy consumption and increasing efficiency across all components of a workload by maximizing the benefits from the provisioned resources and minimizing the total resources required.

Checkpoint

Which category would the EC2 service fall under?

1. Machine Learning
2. Networking
3. Compute
4. Databases

Which of the following is not a pillar of the AWS Well-Architected Framework?

1. Cost Optimization
2. Harmony
3. Security
4. Reliability

Which of the following is an AWS storage service?

1. EC2
2. Lambda
3. Route 53
4. S3

We looked at…

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Next stop: AWS Identity and Access Management